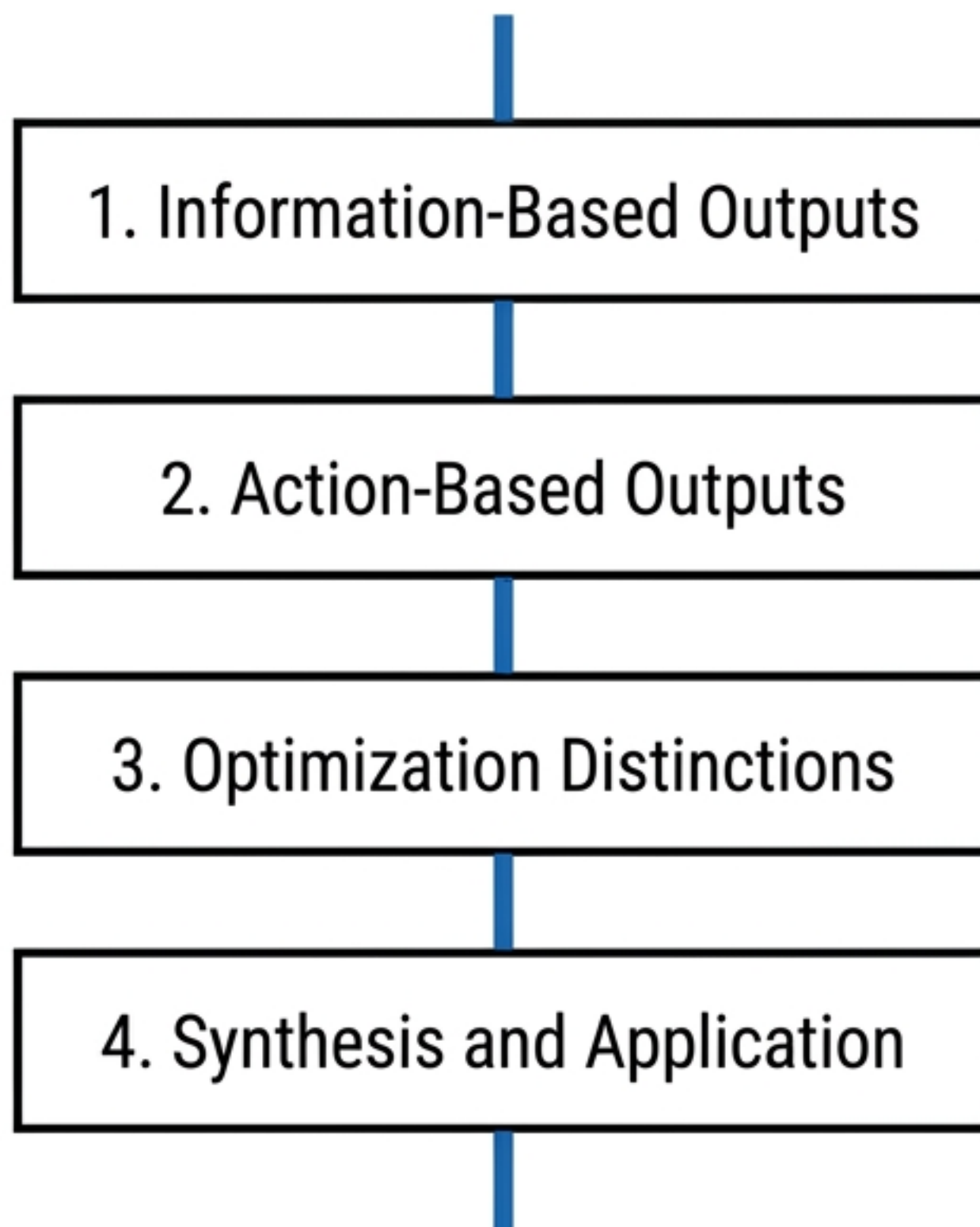


Module 2, Lesson 2

AI 101 for Mechanical Engineers

Different questions need different outputs

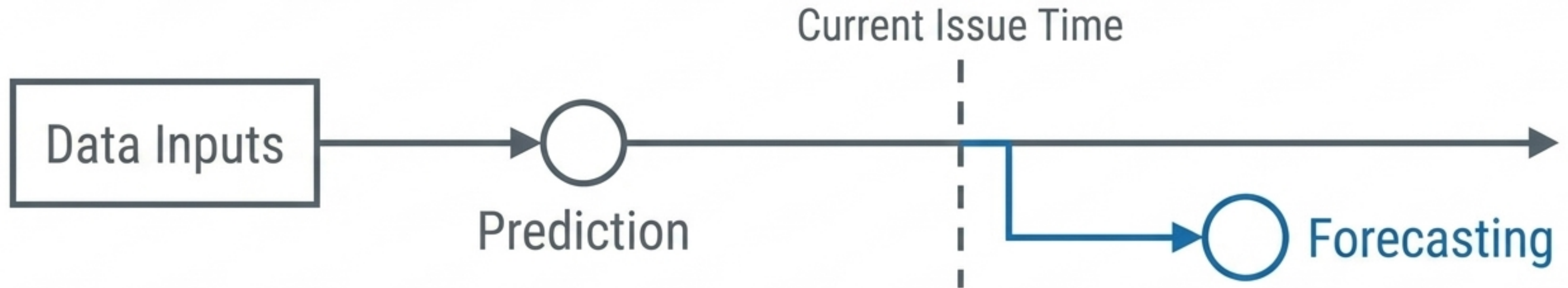
Lesson Structure



1. Information-Based Outputs

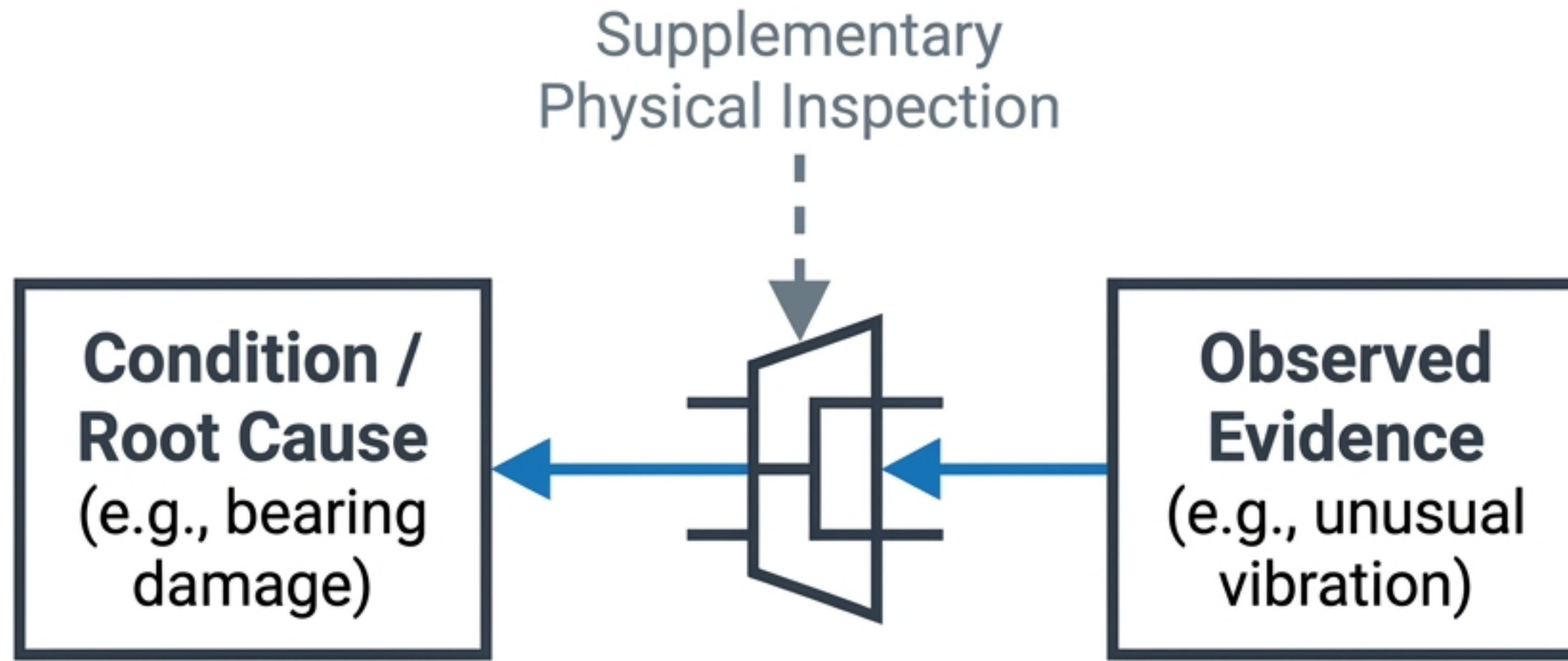
Define outputs that generate information without altering system state.

Prediction and Forecasting



- Estimate unknown outcomes from supplied inputs.
- Predict strictly future physical outcomes.
- Restrict forecasting inputs to present-time data.

Diagnosis

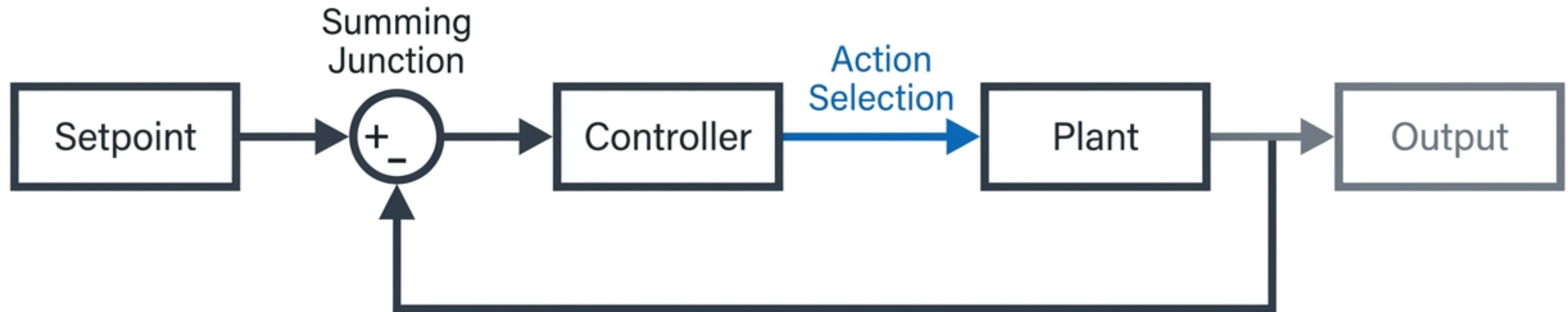


- Identify system conditions from observed evidence.
- Distinguish predictive associations from physical causality.
- Require supplementary evidence for physical confirmation.

2. Action-Based Outputs

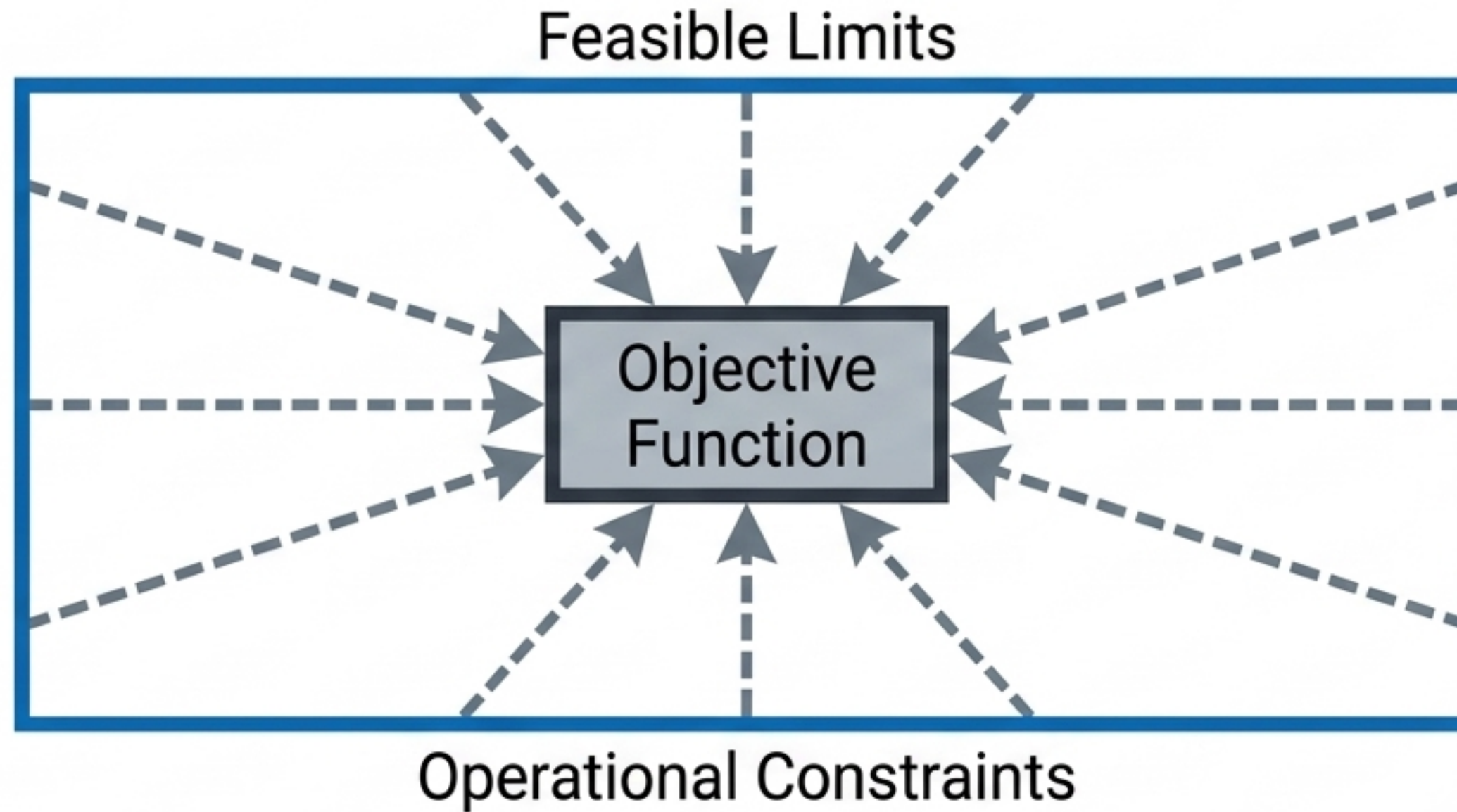
Define outputs that actively regulate or modify system parameters.

Control



- Select actions iteratively over time.
- Regulate dynamic system behaviour.
- Track defined operational system setpoints.

Optimization

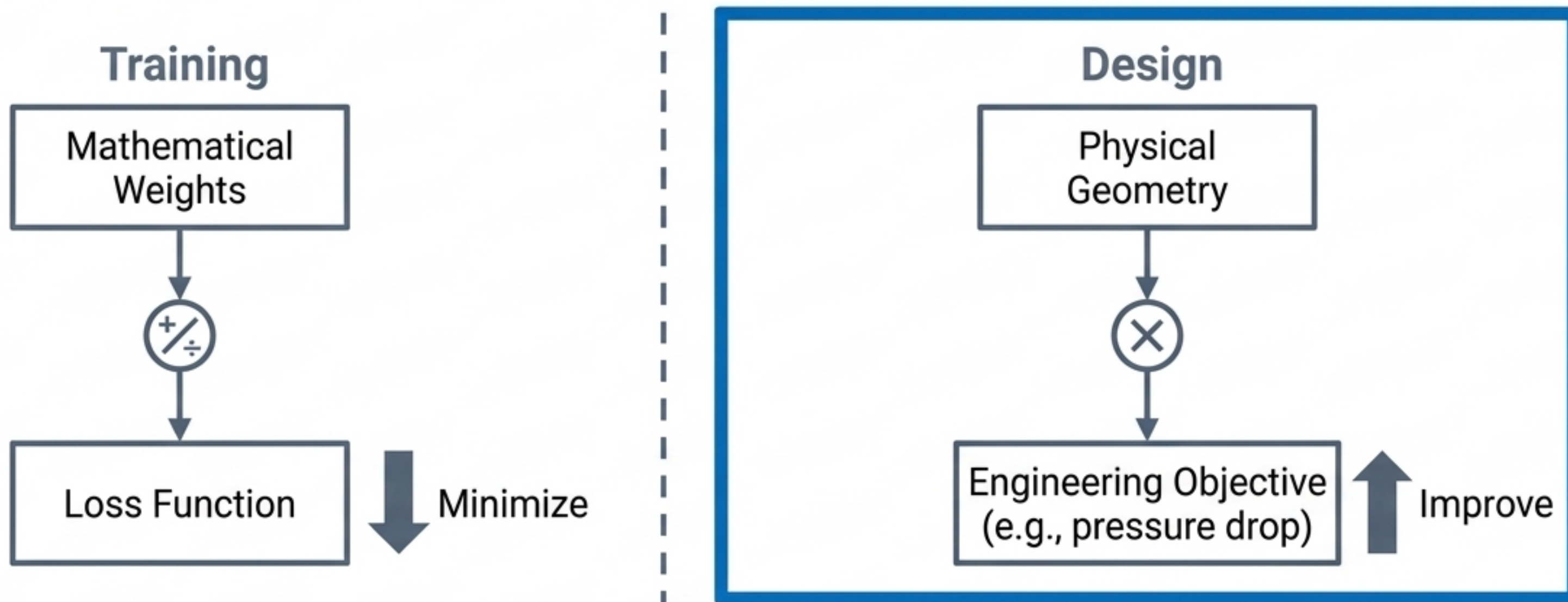


- Search for optimal system configurations.
- Improve a specific objective function.
- Enforce operational constraints and limits.

3. Optimization Distinctions

Separate mathematical model training
from physical system design.

Training vs. Engineering Design



- Distinguish mathematical machine learning from engineering design.
- Training minimizes mathematical loss functions.
- Design improves physical engineering objectives.

4. Synthesis and Application

Map theoretical definitions to
mechanical engineering case studies.

Taxonomy Summary Matrix

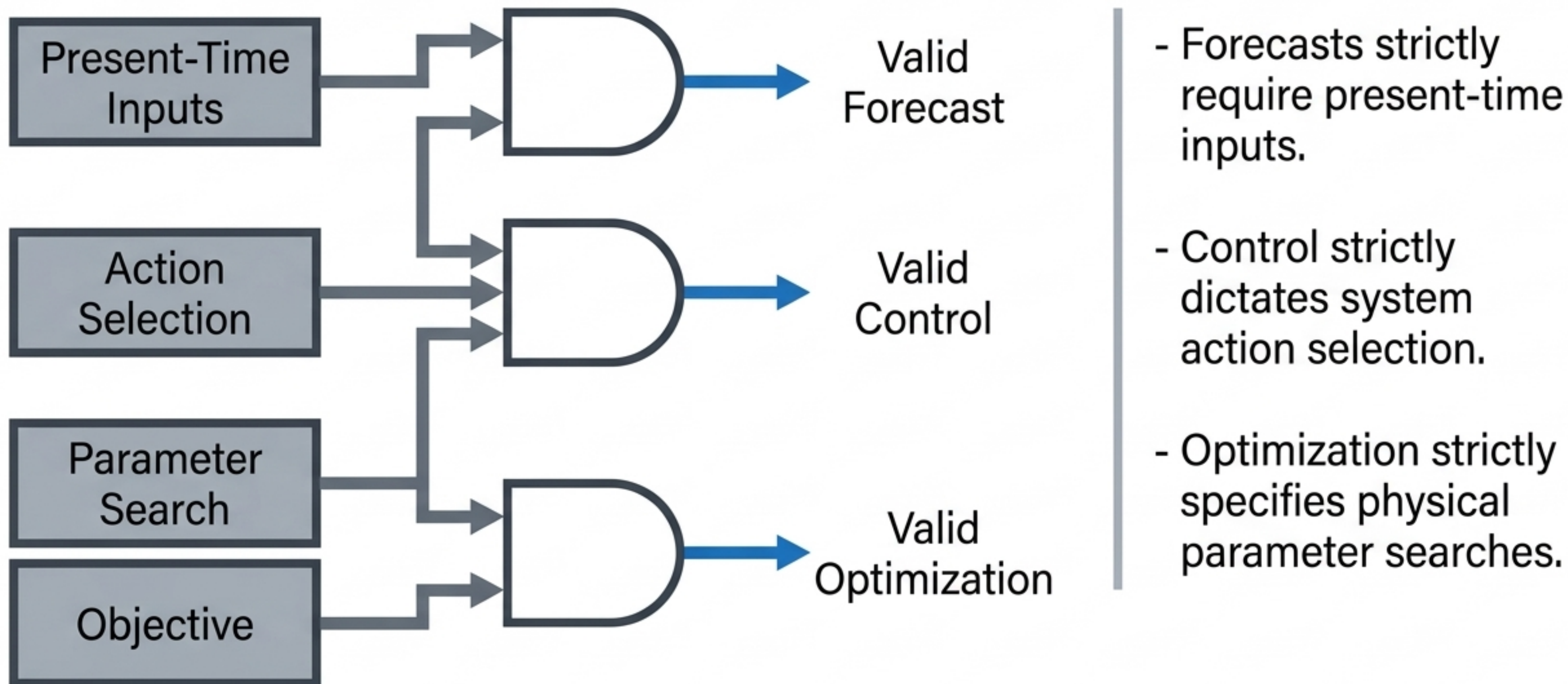
Task	Guiding Question	Mechanical Engineering Example
Prediction	What is the unknown output?	Power at supplied inputs
Forecasting	What happens later?	Tomorrow's PV energy / electricity
Diagnosis	What condition explains evidence?	Confirmed bearing damage
Control	Which action regulates behaviour?	Fan-speed command
Optimization	Which feasible choice improves an objective?	Insulation thickness

Case Study: Pump Operation



- Predicting power output supplies operational estimates.
- Optimization requires defined decision variables.
- Selection demands physical constraint verification.

System Constraints Verification



Key Points

- Information outputs estimate unknowns; action outputs regulate states.
- Mathematical model training fundamentally differs from physical design optimization.
- Prediction isolates variables; optimization and control integrate system parameters.
- Mechanical system design requires defined objectives, constraints, and verification.

